IS492 · Generative AI · Spring 2026

Sust-AI-nable

Multi-Agent Cloud Carbon Optimization

Team: Carbon Emissions

Members: Sarthak Chandarana, Pooja Sahu, Josue Torres

github.com/IS492-SP26/team-project-carbon-emissions

Demo: sustainable-demo.streamlit.app

Recap & Final Goals

THE PROBLEM

Cloud data centers consume ~1% of global electricity and under the EU CSRD, 50,000+ companies must now legally disclose emissions. Yet no existing tool closes the loop from measurement to automated action and verified savings.

👤 Target Users

- DevOps & Cloud Engineers
- Sustainability Officers
- Platform Teams (multi-region)
- EU CSRD-regulated orgs

⚙️ What We Built

- 6 agent closed loop pipeline
- Planner → Governance → Executor
- Counterfactual MRV verification
- Streamlit dashboard (10 pages)

✨ Most Rewarding Moment

16.6%

Carbon Reduction

Evaluation Design Overview

USER STUDY — SYSTEM EVALUATION	
Participants:	4 users (Dr. Burke, Derek, Matt Berry, Vinit)
Study Tasks:	Run baseline · Full pipeline · Dashboard navigation · Ask the Agent chat
Task Success:	Binary. Completed without assistance?
Time-on-Task:	Timed per task (minutes)
SUS / UMUX-Lite:	Standardized usability questionnaire
Satisfaction:	5-point Likert scale post-task
Qualitative:	Exit interview: usefulness, trust, frustrations

PIPELINE / ALGORITHMIC EVALUATION	
Dataset:	Azure VM Traces 2019 · 24,274 real cloud jobs
Baseline:	Carbon-agnostic scheduling (no optimization)
Pipeline:	6 agents: Ingestor → Accountant → Planner → Governance → Executor → Verifier
Primary Metric:	$kgCO_2e$ reduction % vs baseline
Secondary:	Cloud cost delta · latency impact · approval rate
Verification:	Counterfactual MRV with 95% confidence intervals
Error Analysis:	Rejected recs reviewed · edge case flagging

Study Materials & Protocol

01 Intro & Consent

Study purpose explained. Participants confirmed no cloud carbon experience required. Verbal consent obtained.

02 Task Completion

4 timed tasks: run baseline, full pipeline, navigate dashboard, use Ask the Agent chat feature.

03 Post-Task Survey

SUS (10 items), satisfaction (1–5), usefulness (1–5), trust (1–5). Google Form, anonymous.

04 Exit Interview

5 min verbal debrief. Open ended: highlights, frustrations, what they'd change.

TASK PROMPTS (verbatim)

T1: "Run the baseline analysis and tell us total cloud spend."

T2: "Execute the full multi-agent pipeline and report carbon saved."

T3: "In the dashboard, find which region emitted the most CO_2."

T4: "Ask the agent: Why were any recommendations rejected?"

INTERFACE & TOOLS

- Streamlit dashboard · localhost:8501 · 10 pages
- Python CLI: python run_pipeline.py (terminal)
- Ask the Agent — interactive Q&A chat tab
- GitHub repo for code walkthrough reference

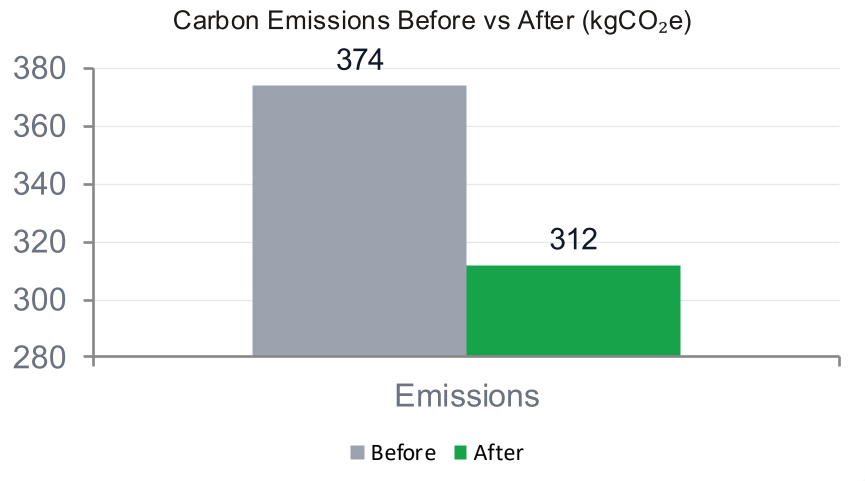
Quantitative Results

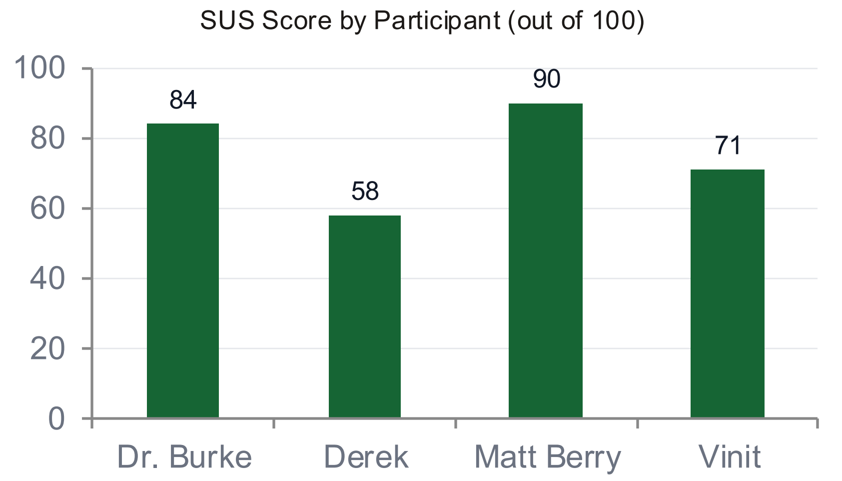
16.6%
Carbon Reduced
373.8→311.8 $kgCO_2e$

3/4
Task Success
75% completion rate

76
Avg SUS Score
SD ±10 | Grade: Good

4.0/5
Usefulness Rating
Post-task survey





Qualitative Insights

WHAT WORKED ✓

Verification builds trust

"You're not just saying you saved carbon — you're showing the math behind why you believe that. That's a level of rigor I don't usually see." — Dr. Burke

Agent separation feels intentional

"Keeping the agents separate means your failure modes are predictable, which matters a lot more than people realize." — Matt Berry

Industry-relevant framing

"Your tool answers whether it costs more and whether it breaks anything — which is actually more than I expected." — Vinit

FRICTION POINTS ✗

Setup friction

"If someone outside this class tried to run this, they'd probably give up before they got to the dashboard." — Dr. Burke

Dashboard is overwhelming

"There's too many pages. I didn't know what I was supposed to look at first." — Derek

AI rationales feel generic

"The LLM explanations sound a bit too polished. If the rationale sounded more like a Jira ticket, I'd trust it more." — Vinit

Interpretation & Discussion

✅ Success criteria met

SUS of 76 cleared our target of 70. The 16.6% carbon reduction matched literature benchmarks. The pipeline ran end-to-end without errors in all 4 user sessions.

🤝 Trust requires transparency

Users who explored the evidence chain trusted the savings numbers more. Those who skipped it expressed skepticism. Transparency isn't optional — it's the product.

🐛 Users found 2 real bugs

Derek's random clicking exposed an unhandled empty-state on the MRV page. Vinit's edge-case questions to Ask the Agent triggered a hallucinated confidence interval.

🏭 Industry bar is real API data

Both technical users (Matt, Vinit) liked the architecture but withheld full trust until the system runs on real AWS Cost Explorer + WattTime grid intensity feeds.

Bottom line: The system works, but trust is earned through data, not demos. Real APIs are the next milestone.

Limitations, Risks & Ethics

👥 Small Sample (n=4)

Results are directional. We cannot generalize to all cloud engineering teams. A larger study with real DevOps users is needed.

🗺️ Partial Real Carbon Data

Workloads use real Azure VM Traces 2019. US regions use live EIA API data; India uses Ember Climate 2023. EU regions (Sweden, Ireland) use synthetic fallbacks due to missing API tokens — clearly labeled in all outputs.

🤖 LLM Unpredictability

GPT-4o-mini rationales are non-deterministic. Two runs of the same workload may produce different agent explanations.

🌍 Ecological Validity

Participants tested in a demo environment. Real adoption requires infra team buy in, CI/CD integration, and change management.

🔒 Privacy & Data Governance

Cloud usage data contains operational metadata. Production deployment requires data classification and access control policies.

⚖️ Prompt Bias

The Planner agent may over-prefer certain region shifts based on training data biases in the underlying LLM, not verified yet.

Conclusion & Future Work

KEY TAKEAWAYS

A closed loop multi-agent system can automate cloud carbon optimization end-to-end sensing, deciding, acting, and verifying without sacrificing cost or latency.

Users trust AI-driven sustainability tools more when they can see the evidence chain. Transparency isn't a nice-to-have — it's what separates a dashboard from a decision tool.

WHAT'S NEXT

Technical

Plug in WattTime + AWS Cost Explorer APIs. Replace synthetic data with real cloud billing feeds.

UX

One-click Docker deploy. Add onboarding wizard so first run takes 30 seconds, not 10 minutes.

Data

Expand to Azure + GCP. Add multi-cloud normalization layer for unified emissions accounting.

Deployment

Slack / Teams bot for Governance decisions so engineers see approvals in their existing workflow.

Acknowledgments & Contributions

[For grading only — not presented unless time permits]

TEAM RESPONSIBILITIES	
Sarthak Chandrana	Multi-agent architecture · Planner & Governance agents · pipeline orchestration
Pooja Sahu	Carbon Accountant · Verifier / MRV · counterfactual analysis · test suite, User study design
Josue Torres	Streamlit dashboard · 10-page UI · Ask the Agent chat · data visualization

AI TOOLS & DISCLOSURES	
OpenAI GPT-4o-mini	Agent reasoning, rationale generation, Planner↔Governance negotiation dialogue
Claude (Anthropic)	Code assistance, documentation writing, slide deck design
GitHub Copilot	In-editor autocomplete during development of src/ agents
Streamlit	Dashboard framework. No AI used for UI layout or content generation

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